Surya Manoj Sanu

Scientific Machine Learning · Differentiable Simulation · Numerical Optimization

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Applied AI researcher with five years of experience in differentiable simulation and design optimization, following a path from mechanical engineering through nanoscale materials science into scientific machine learning. Works through rigorous foundational analysis, unifying fragmented literature and isolating what actually drives performance in black-box systems, then communicates that understanding in a form others can build on.

Education

Ph.D., Machine Learning and Computational Design 12/2021 – 2026 (thesis submitted)
Delft AI Lab, TU Delft, The Netherlands — *Finding Structure in Neural Landscapes: Reparameterization, Hypergradients, and Bilevel Topology Optimization*

M.Sc., Materials Science and Engineering 09/2019 – 11/2021
TU Delft, The Netherlands — *Cum Laude*, thesis graded 9.0
Specialization in Nanomaterials and Nanoscience

B.Tech., Mechanical Engineering 08/2013 – 07/2017
NIT Calicut, India — Grade 8.3/10 (All-India Rank 2601, ~1.4M candidates)

Selected publications

Sanu, S.M., et al. Bilevel Optimization of Topology and Hyperparameters (BOTH). *arXiv*, 2026.
Sanu, S.M., et al. Leveraging automatic differentiation in modern machine learning frameworks for (neural) topology optimization. *Struct Multidisc Optim*, 2026.
Sanu, S.M., Aragón, A.M., Bessa, M.A. Neural topology optimization: the good, the bad, and the ugly. *Struct Multidisc Optim*, 68, 213, 2025.

Research Experience

During Ph.D.: 2021 – Present

- Authored an educational paper and open-source codebase (JAX/PyTorch) demonstrating the integration of automatic differentiation into design optimization workflows, unifying adjoint analysis and iterative solver sensitivities under the Implicit Function Theorem.
- Unified neural parameterization workflows described under varying terminology across the literature, identifying inconsistent evaluation metrics that prevented fair comparison between studies and motivating a systematic benchmark of ≈1,000 reproducible runs spanning neural architectures (CNNs, neural fields), optimizers, boundary conditions, and mesh resolutions.
- Developed diagnostic tools for loss-landscape analysis of the high-dimensional neural optimization, opening up the black box by tracing failures to limited network expressivity and ill-conditioning and revealing high sensitivity to hyperparameters.
- Developed **BOTH**, a framework combining bilevel optimization, automatic differentiation and machine learning to tune hyperparameters jointly with neural design optimization via unrolled hypergradients through the physics-based training loop: it converges from initializations where fixed-hyperparameter baselines stalled, outperforms Bayesian optimization in low dimensions, and scales to ≈10,000 hyperparameters, where Bayesian optimization becomes intractable. The framework also supports other bilevel tasks such as meta-learning and neural architecture search.
- Spearheaded the selection of a software platform for the research group's differentiable simulation research, evaluating JAX, Julia, MFEM and FEniCS on automatic differentiation, GPU support, ML

integration and extensibility, and co-architected the resulting JIT-compiled GPU framework, CRAX (up to 3M design variables), which became the shared codebase and enabled research on meta-learning, nonlinear material models, and embedded neural surrogates.

Master Thesis: Exploring the optimization process in neurally reparameterized topology optimization 2020–2021

- Tested the assumption that L-BFGS outperforms standard ML optimizers in neural topology optimization because it captures curvature, implementing it from scratch in TensorFlow and adopting diagnostics from the ML literature (gradient dynamics, loss-landscape geometry, Hessian-vector products), revealing that line-search step-size selection, rather than curvature accuracy, drove most of the advantage.

Teaching & Scientific Communication

- **Journal club:** Presented and critically dissected state-of-the-art and seminal AI-for-science papers in a weekly group journal club, assessing their claims, assumptions and limitations.
- **Conference talks:** U.S. National Congress on Computational Mechanics (2023); World Congress of Structural and Multidisciplinary Optimization (2023, 2025).
- **Lecturing (TU Delft & Brown University):** Delivered Master's-level guest lectures on gradient-based optimization and automatic differentiation; designed theoretical coursework on Bayesian ML for data-driven design; teaching assistant for Bayesian ML and computational materials science courses.
- **Master's thesis supervision (ongoing):** Directing research on learning flexible, boundary-conditioned geometric parameterizations to improve local-minima identification in non-convex landscapes, and on accelerating convergence and enforcing sparsity via adaptive neural network pruning.
- **Bachelor capstone supervision:** Initiated and managed projects on generative diffusion models for super-resolved design generation and neural ensembles for memory-efficient representations.

Skills & Languages

Applied mathematics: Numerical optimization, finite element solution of PDEs, nonlinear equation solvers, implicit differentiation
Machine learning: Automatic differentiation, bilevel optimization and hypergradients, CNNs and encoder–decoder (U-Net) architectures, neural fields, hyperparameter optimization, scientific machine learning, Transformers, physics-informed learning
Software: Python, JAX, PyTorch, TensorFlow, NumPy/SciPy, PETSc, NVIDIA AMGX, Matlab, Git, Linux/Bash, HPC (Slurm)
AI-assisted development: Claude Code, GitHub Copilot
Languages: English (fluent), Dutch (conversational)

Activities

Organizer, Poster fair, ELLIS ML symposium, TU Delft 2026 – Present
PhD Representative, TU Delft MSE 2021 – 2022
Education Commissioner, Tubalkain (Student Association) 2019 – 2020
Board of Studies Representative, TU Delft MSE 2019 – 2021

Interests

Reading fiction, squash at an amateur level, inventing (con)fusion dishes, and long brunches with friends